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HOT SPOTS ON NEUTRON STARS: THE NEAR-FIELD GRAVITATIONAL LENS

K. R. PECHENICK,[1] C. FTACLAS, AND J. M. COHEN[1]
University of Pennsylvania
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ABSTRACT

We present a simple model for including gravity in the light curves of slowly rotating neutron stars whose emission is confined to circular antipodal polar caps. Both thermal and near thermal cases are considered. We find that for hot polar caps and typical neutron star parameters the light curves are essentially flat as a result of gravitational effects. If, however, photon emission is beamed, structure is, in general, restored to the light curve, depending on the nature of the beaming. For the most relativistic neutron stars we find that there can be sharp peaks in the light curve induced solely by the gravitational lensing effect of the star. We also present results for a single hot spot. In an appendix we derive the tidal deformation of an initially circular cone of light rays.

Subject headings: gravitation — relativity — stars: neutron

I. INTRODUCTION

Neutron stars represent the most relativistic stellar surfaces accessible to observation. Yet while a great variety of effects have been observed, inferred, or proposed as being associated with these objects (Manchester and Taylor 1977; Smith 1977; Irvine 1978), gravitational effects, apart from obvious redshift considerations and the binary pulsar (Hulse and Taylor 1975), have heretofore remained essentially unexplored.

In this communication we begin a program to study the effects of gravity on the light curves of neutron stars by considering a rotating star whose luminosity originates solely in restricted polar regions, either single or antipodal. Our results would be applicable to thermal emission arising from accretion (cf. Helfand, Chanan, and Novick 1980; Pines 1980) or single hot spots that might arise on the surface. Most recently Wang and Welter (1981, hereafter WW) have analyzed X-ray binary light curves using flat-space models. The current work can be considered a generalization of their analysis. In § II the basic parameters of our model are presented and photon orbits are discussed. In § III the observed brightness of a polar cap is calculated and applied to the generation of light curves in § IV. Our results are presented in § V and discussed in § VI. In an appendix we derive the tidal deformation of an initially circular cone of light rays.

II. BASIC GEOMETRY AND PHOTON ORBITS

The metric outside the star is taken to be the Schwarzschild metric with mass M and radial coordinate r. The surface of the star is at $r = R$ with $R/2M > 1.5$, so the star is larger than its "photon sphere." An anonymous referee has pointed out that neutron stars could, in principle, be more strongly relativistic (Weinberg 1972, p. 334). As above, we will assume $G = c = 1$ throughout. The luminous emission of the star is assumed to originate at $r = R$ with constant intensity I over single or antipodal polar caps of angular radius α. If δ is an angular variable measured relative to the normal to the stellar surface, then I may depend on δ, so that the emission is assumed homogeneous but not necessarily isotropic. (Note: Isotropic = independent of δ = "Lambert's law".) If I depends on δ, we say the radiation is "beamed." The approach taken in the next two sections is based on the treatment by Misner, Thorne, and Wheeler (1973) and Matese and Whitmire (1978).

The observer is taken to be stationary at $r = r_0$ where $r_0 \to \infty$. Since neutron stars are "slowly rotating" from the point of view of general relativity, we assume the brightness of the rotating star to have the value it would have at that instant if the star were stationary; we do not use the Kerr metric and we neglect the "dragging of inertial frames" (Brill and Cohen 1966; Cohen and Brill 1968). We also assume that the region $r < R$ is totally opaque and that apart from beaming conditions introduced at the caps, the region $r > R$ is totally transparent, so we ignore effects on photon propagation due to physical conditions other than gravity in the region surrounding the star. With these assumptions the observed brightness l is the integral of the intensity at the observer over the solid angle subtended in the observer's sky by the luminous regions on the star. If there is no beaming, then, as in flat space, I is a constant in the observer's sky (see the Appendix) and l is directly pro-

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portional to Ω, the subtended solid angle. If there are several images, they must all be summed over; and if there is beaming, then l is an integral over all images of the cap. Since null geodesics do not depend on photon frequency, the frequency dependence of the emitted radiation is not relevant to the problem assuming a perfect detector.

Consider an observer (in an arbitrary curved space-time) who sees two photons simultaneously. Let the four-velocity of the observer be u^μ, and let v^μ and w^μ be the four-velocities of the two photons. The observer sees an angle ξ between the directions of the two photons, where

$$\cos\xi = 1 + (v^\mu w_\mu)[(u^\nu v_\nu)(u^\sigma w_\sigma)]^{-1}. \quad (2.1)$$

This follows since in the observer's rest space with the metric $h_{\alpha\beta} = g_{\alpha\beta} + u_\alpha u_\beta$ the photons appear as the three-vectors $v'^\mu = h^{\mu\alpha} v_\alpha$ and $w'^\mu = h^{\mu\alpha} w_\alpha$ (Ellis 1971; Ftaclas and Cohen 1980). Equation (2.1) derives from the inner product of v'^μ and w'^μ in this 3-space.

For the Schwarzschild metric, a photon in the "equatorial plane" ($\theta = \pi/2$) is described by the impact parameter b, which determines the precise shape of the orbit. For such a photon, λ can be chosen so that (Misner, Thorne, and Wheeler 1973, p. 674)

$$v^0 \equiv \frac{dt}{d\lambda} = (1-2M/r)^{-1},$$

$$v^1 \equiv \frac{dr}{d\lambda} = [1 - b^2(1-2M/r)/r^2]^{1/2},$$

$$v^2 \equiv \frac{d\theta}{d\lambda} = 0,$$

$$v^3 \equiv \frac{d\phi}{d\lambda} = br^{-2}. \quad (2.2)$$

A particular case of equation (2.2) is a radially moving photon, for which $b = 0$. Letting w^μ represent the four-velocity of this radial photon, we have

$$w^\mu = [(1-2M/r)^{-1}, 1, 0, 0]. \quad (2.3)$$

The four-velocity of a stationary observer in the Schwarzschild metric is

$$u^\mu = [(1-2M/r)^{-1/2}, 0, 0, 0]. \quad (2.4)$$

It follows from equation (2.1) that the angle in the observer's sky between v^μ and w^μ is given by

$$\cos\xi = [1 - (1-2M/r_0)b^2/r_0^2]^{1/2}$$

so that

$$\sin\xi = (b/r_0)(1-2M/r_0)^{1/2}, \quad (2.5)$$

which in the limit $r_0 \to \infty$ can be written

$$\xi = b/r_0. \quad (2.6)$$

Let us now suppose that the photons are emitted from the surface ($r = R$) of an opaque sphere (the neutron star). The photon described by equation (2.2) obeys (Misner, Thorne, and Wheeler 1973, p. 673)

$$\left(\frac{1}{r^2}\frac{dr}{d\phi}\right)^2 + B^{-2}(r) = \frac{1}{b^2}, \quad (2.7)$$

where

$$B^{-2}(r) = (1-2M/r)/r^2. \quad (2.8)$$

Because of equation (2.7), a photon at r must have an impact parameter b such that

$$b^{-2} - B^{-2}(r) \geq 0. \quad (2.9)$$

The function $B^{-2}(r)$ has a maximum at $r = 3M$ and decreases for $r > 3M$. Thus, if the sphere has $R > 3M$, then a photon emitted from the surface and reaching the observer must have an impact parameter $b \leq b_{\max}$, where

$$b_{\max} = R(1-2M/R)^{-1/2}. \quad (2.10)$$

Substituting $u = M/r$, and $u_b = M/b$, equation (2.7) becomes

$$\left(\frac{du}{d\phi}\right)^2 + (1-2u)u^2 = u_b^2. \quad (2.11)$$

If the photon travels from $r = R$ to $r = r_0 \to \infty$, in the equatorial plane, then

$$\Delta\phi = \int_0^{M/R} [u_b^2 - (1-2u)u^2]^{-1/2}\, du. \quad (2.12)$$

The maximum possible value of $\Delta\phi$ occurs when $b = b_{\max}$. From equations (2.10) and (2.12),

$$(\Delta\phi)_{\max} = \int_0^{M/R} [(1-2M/R)(M/R)^2 - (1-2u)u^2]^{-1/2}\, du. \quad (2.13)$$

Equations (2.11) to (2.13) were derived for the equatorial plane, which is no different from any other "plane" through the origin.

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For $M=0$ (flat spacetime), $(\Delta\phi)_{\max}=90°$ and exactly half of the surface of the star is visible. For $R/2M \approx 1.76$, $(\Delta\phi)_{\max}=180°$ and the entire surface of the star is visible; the point at 180° appears as the circular boundary of the disk, whose angular radius is $1.52R/r_0$. As $R/2M$ approaches 1.5 from above, $(\Delta\phi)_{\max}$ approaches infinity. For $R/2M=1.6$, for instance, $(\Delta\phi)_{\max}=227°$, so that all points can be seen at least once and those between 133° and 180° to the observer's line of sight can be seen twice. The point at 180° appears as a circle whose angular radius is $1.621R/r_0$, while the boundary of the entire disk is a circle with angular radius $1.633R/r_0$. In general, the angular radius of the observed disk is given by equation (2.6) with $b=b_{\max}$.

III. THE OBSERVED BRIGHTNESS OF A POLAR CAP

We now choose our coordinates so that the observer is on the positive z-axis, at $r=r_0\to\infty$. The surface of the star is described by angular spherical coordinates, θ and ϕ, where θ corresponds to $\Delta\phi$ in equation (2.12). We use primed coordinates, θ' and ϕ', for the observer's sky, where θ' corresponds to ξ in equations (2.5) and (2.6). The element of solid angle in the observer's sky is

$$d\Omega' = \sin\theta' d\theta' d\phi'; \tag{3.1}$$

or, since θ' is infinitesimally small,

$$d\Omega' = \theta' d\theta' d\phi'. \tag{3.2}$$

We consider a polar cap of angular radius α centered at $\theta=\theta_0$ and define a function $h(\theta;\alpha,\theta_0)$ as the range of ϕ included in the "one-dimensional slice" at θ of the polar cap (see Fig. 1). In order to include multiple images, we allow θ to range from 0 to $(\Delta\phi)_{\max}$, although we define θ_0 to be between 0 and 180°. Since $\phi=\phi'$ for corresponding points of the image and the surface of the star, we can write

$$d\Omega' = h(\theta;\alpha,\theta_0)\theta' d\theta'. \tag{3.3}$$

[For instance, if

$$\theta_0+\alpha \le (\Delta\phi)_{\max} \le 180°$$

and

$$\theta_0-\alpha \ge 0,$$

then

$$h(\theta;\alpha,\theta_0) = 2\arccos\left(\frac{\cos\alpha-\cos\theta_0\cos\theta}{\sin\theta_0\sin\theta}\right) \tag{3.4}$$

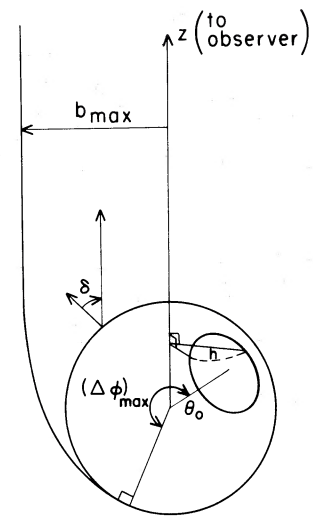

FIG. 1.—The geometry used to determine the brightness of a single cap.

for $\theta_0-\alpha\le\theta\le\theta_0+\alpha$;

$$h(\theta;\alpha,\theta_0)=0 \text{ for } \theta \text{ outside the range } \theta_0\pm\alpha].$$

In equation (3.3), θ is a function of θ' given by

$$\theta = \int_0^{M/R}\left[(M/b)^2-(1-2u)u^2\right]^{-1/2}du \tag{3.5}$$

with

$$b = r_0\theta'. \tag{3.6}$$

Letting

$$x = b/M, \tag{3.7}$$

equation (3.3) becomes

$$d\Omega' = (M/r_0)^2 h(\theta;\alpha,\theta_0)\,x\,dx, \tag{3.8}$$

where

$$\theta = \int_0^{M/R}\left[x^{-2}-(1-2u)u^2\right]^{-1/2}du. \tag{3.9}$$

From the discussion in § II,

$$dl = I(r_0,\Omega')\,d\Omega' = (1-2M/R)^2 I(R,\Omega')\,d\Omega' \tag{3.10}$$

Because of the homogeneity of the radiation, I is independent of ϕ. However, I can depend on δ, which becomes a function of θ when we impose the require-

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ment that the photons under consideration must reach the observer. Thus, I is a function of x. The function $I(\delta)$ is assumed to be given, so we only need to determine $\delta(x)$. From equation (2.5) with $\xi \to \delta$ and $r_0 \to R$, we have

$$\sin\delta = (b/R)(1-2M/R)^{1/2}. \tag{3.11}$$

Defining $x_{\max} = b_{\max}/M$, we have from equation (2.10)

$$x_{\max} = (R/M)(1-2M/R)^{-1/2}. \tag{3.12}$$

Thus,

$$\delta = \arcsin(x/x_{\max}). \tag{3.13}$$

From equations (3.8), (3.10), and (3.13),

$$dl = (1-2M/R)^2 I[\arcsin(x/x_{\max})] \times (M/r_0)^2 h(\theta;\alpha,\theta_0)\, x\, dx. \tag{3.14}$$

Equation (3.14) must be integrated over the range $0 \le x \le x_{\max}$, with $\theta(x)$ given by equation (3.9). The total brightness of one polar cap may be written as

$$l = (R/r_0)^2 I_0 A(\theta_0; f, M/R, \alpha) \tag{3.15}$$

in which $I(\delta) = I_0 f(\delta)$, and

$$A(\theta_0; f, M/R, \alpha) = \left(1-\frac{2M}{R}\right)^2\left(\frac{M}{R}\right)^2 \times \int_0^{x_{\max}} f(x) h[x;\alpha,\theta_0]\, x\, dx. \tag{3.16}$$

The quantity A includes only one polar cap. We define a related quantity, B, which is proportional to the combined brightness of two polar caps with their centers at antipodal points:

$$B(\theta_0; f, M/R, \alpha) = A\left(\theta_0; f, \frac{M}{R}, \alpha\right) + A\left(\pi-\theta_0; f, \frac{M}{R}, \alpha\right). \tag{3.17}$$

In equation (3.17), no information is lost if θ_0 is restricted to the range from 0° to 90°.

IV. LIGHT CURVES

To obtain light curves (for two polar caps) predicted by our model, it is necessary to determine θ_0 as a function of time, where θ_0 ($0° \le \theta_0 \le 90°$) is the angle from the observer to the center of the nearer polar cap. We can then use the graphs of $B(\theta_0)$ to plot graphs of l versus time.

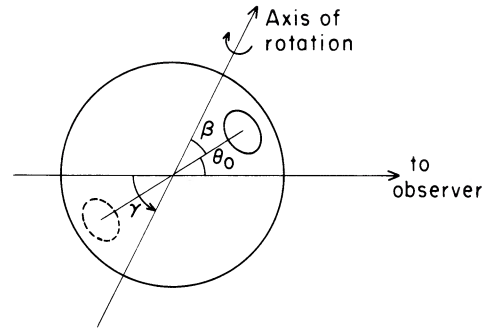

FIG. 2.—The relation of the polar cap axis and the axis of rotation to the observer's line of sight.

If β is the angle between the axis of rotation of the star and the line joining the centers of the two polar caps, and γ the angle between the axis of rotation and the line of sight (see Fig. 2), then

$$\theta_0 = \arccos|\sin\beta\sin\gamma\cos\omega t + \cos\beta\cos\gamma|, \tag{4.1}$$

where ω is the angular velocity of the star. With no loss of generality, β and γ may be restricted to the range from 0° to 90°. It follows from equation (4.1) that the second half of the light curve ($180° \le \omega t \le 360°$) is a mirror image of the first half ($0° \le \omega t \le 180°$) and that each graph of B versus θ_0 is actually a light curve for one-fourth of a rotation, for the special case of $\beta = \gamma = 90°$. For this case, as well as when only one of β or γ is 90°, the light curve repeats itself after half a rotation and the second quarter of the light curve is a mirror image of the first.

We emphasize the fact that a graph of $B(\theta_0)$, with $0 \le \theta_0 \le 90°$, contains enough information to plot any light curve for two polar caps. Given rotational symmetry about the line of sight, any configuration achieved with arbitrary values of β and γ is also achieved in the graph of $B(\theta_0)$. Changing β and γ generally limits the proximity of either cap to $\theta = 0$. For this reason, the light curves with $\beta = \gamma = 90°$ exhibit the largest possible range of brightness.

V. RESULTS

In generating our results we have investigated extreme as well as representative cases. For the parameter $R/2M$ we have considered values from $R/2M = 1.6$, just outside the photon sphere, corresponding to the most massive neutron stars, through $R/2M = 2.5$–4.0 corresponding to nominal neutron stars, to $R/2M = \infty$, corresponding to flat spacetime. Our nominal range is

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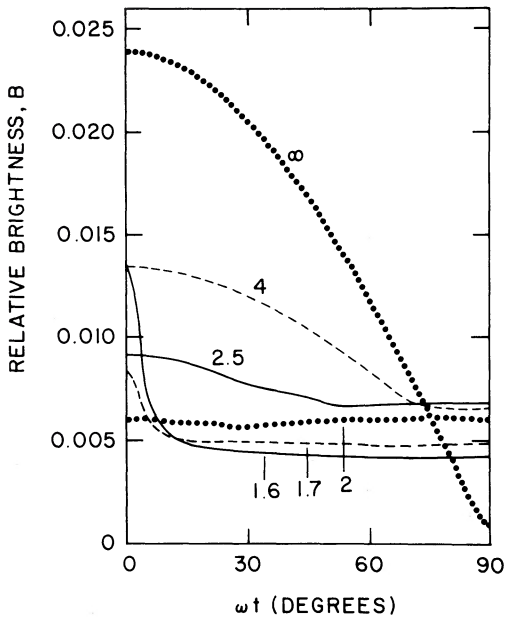

FIG. 3.—Relative brightness B (two polar caps) vs. ωt for α (angular radius of cap) $= 5°$, $f(\delta) = 1$ (no beaming), $\beta = \gamma = 90°$. Numbers denote values of $R/2M$.

arrived at by noting that neutron star masses as measured in binary systems (Taylor, Fowler, and McCulloch 1979; Bahcall 1978) indicate a mass of $\sim 1.4\ M_\odot$ and theoretical models of neutron stars give radii of 10–16 km (see Pines 1980 for a review of this work). Although our primary concern is isotropic emission $[f(\delta) = 1]$, we have also considered cases where physical conditions at the caps have either enhanced $[f(\delta) = \cos\delta]$ or suppressed $[f(\delta) = \sin\delta]$ emission in the radial direction (it is a convenient coincidence that these cases were also chosen by WW). For the sake of completeness and insight into the nature of the processes involved, we have also included the light curves due to single caps. These could represent possible stellar hot spots. Since we want to eliminate model and distance dependence, our curves are actually plots of $A(\theta_0)$ (one cap) or $B(\theta_0)$ (two caps) versus ωt (l for these cases is defined in eq. [3.15] or [3.17], respectively). Because of this simplification the amplitudes of curves with different values of $R/2M$ should not be directly compared to each other, since we have in effect assumed that all stars have the same value of R/r_0 (equivalently one can consider the curves as arising from a star whose radius is kept constant and whose mass is increased, in effect, "turning-on" gravity).

In Figure 3 we show the light curves for one-fourth of a rotation due to two polar caps with $\alpha = 5°$, $\beta = \gamma = 90°$, and assuming no beaming $[f(\delta) = 1]$. These curves have several interesting features. We note first that for the flat space case there is a change of ~ 2.7 mag between minimum and maximum light. As configurations become more relativistic, however, the light curve begins to flatten and by $R/2M = 2.5$ the maximum observable change is only 0.2 mag. By $R/2M = 2$ the light curve is effectively flat, and gravity has acted to eliminate totally the expected flat space behavior. For more relativistic configurations a new feature appears in the form of a spike near $\omega t = 0°$ whose width at half-maximum is approximately the angular diameter of the caps (10°). For the most relativistic case this feature represents a jump of 1.2 mag over the essentially flat continuum. This demonstrates that gravity alone is ca-

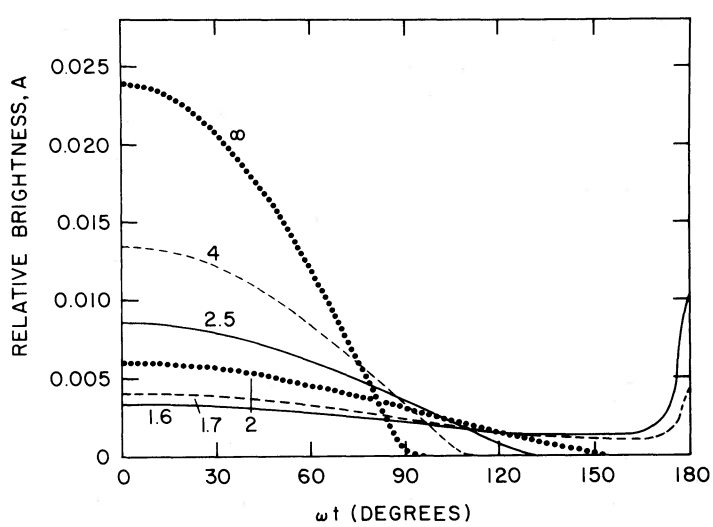

FIG. 4.—Relative brightness A (one polar cap) vs. ωt for $\alpha = 5°$, $f(\delta) = 1$, $\beta = \gamma = 90°$. Numbers denote values of $R/2M$.

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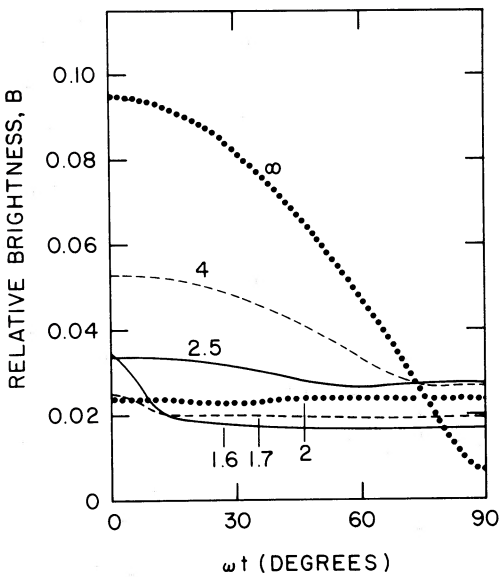

FIG. 5

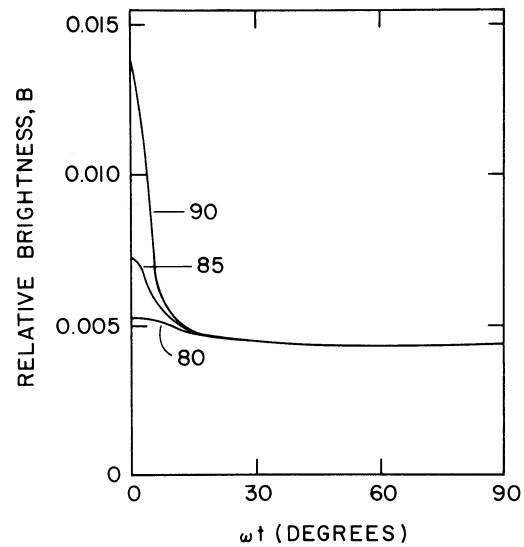

FIG. 6

FIG. 5.—Relative brightness B (two polar caps) vs. ωt for $\alpha = 10°$, $f(\delta) = 1$, $\beta = \gamma = 90°$. Numbers denote values of $R/2M$.

FIG. 6.—Relative brightness B (two polar caps) vs. ωt for $\alpha = 5°$, $f(\delta) = 1$, $\beta = 90°$, $R/2M = 1.6$. Numbers denote values of γ in degrees.

pable of producing strongly beamed radiation even for isotropically radiating polar caps. To illustrate the origin of this beaming we give in Figure 4 the light curves for one-half rotation due to a single 5° polar cap ($\beta = \gamma = 90°$). For flat space, as expected, the light curve goes to zero at $\omega t = 95°$, but for more relativistic configurations the cap is visible beyond $\omega t = 95°$, and for configurations with $R/2M$ slightly smaller than 1.8 the cap is always visible. The sharp peak in Figure 3 is actually due to the gravitational lensing effect of the star on the cap at $\theta_0 = 180°$. It arises from photons emitted near the tangential plane and bent through large angles [recall that for $R/2M = 1.6$, $(\Delta\phi)_{\max} = 227°$]. The region near $\theta_0 = 180°$ appears as a ring whose apparent brightness exceeds that of a comparable region at $\theta_0 = 0°$. The peak will have a width comparable to the angular diameter of the cap; and the smaller the cap, the more pronounced is its height above the continuum. The ratio of its amplitude to continuum amplitude varies like $\alpha^{-0.52}$ based on several computer runs with various cap sizes. These points are illustrated in Figure 5, where we have repeated the curves of Figure 3 using $\alpha = 10°$.

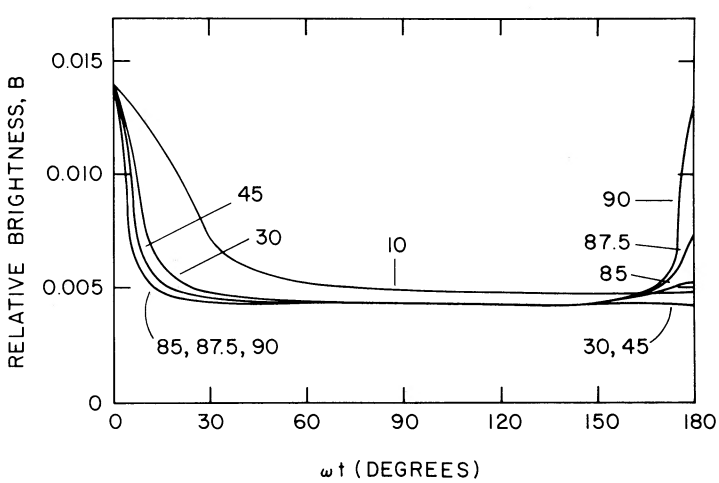

FIG. 7.—Relative brightness B (two polar caps) vs. ωt for $\alpha = 5°$, $f(\delta) = 1$, $R/2M = 1.6$, $\beta = \gamma = x$. Numbers denote values of x in degrees.

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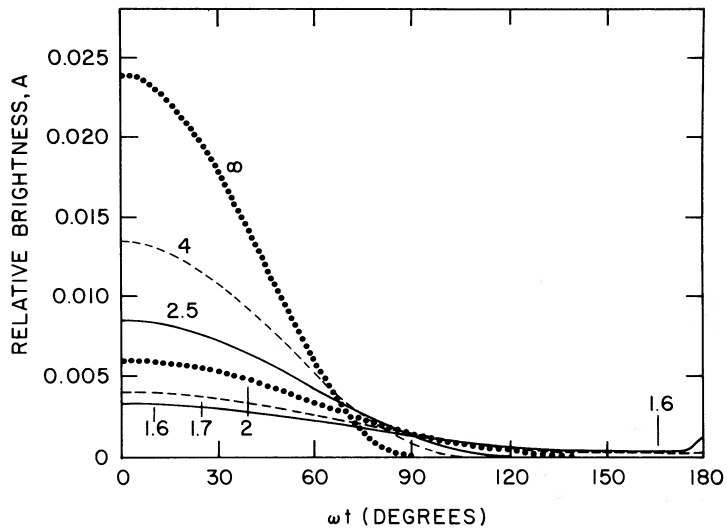

FIG. 8.—Relative brightness A (one polar cap) vs. ωt for $\alpha = 5°$, $f(\delta) = \cos\delta$, $\beta = \gamma = 90°$. Numbers denote values of $R/2M$.

While the general features of these curves are as in Figure 3, the peak for $R/2M = 1.6$ is now broader and only 0.73 mag above the continuum. Since there is very little qualitative difference between Figures 3 and 5, we will confine further discussion to 5° polar caps. In order to further study the peak occurring for very relativistic configurations, we present in Figures 6 and 7 light curves for 5° caps and $R/2M = 1.6$ but various possibilities for β and γ. In Figure 4 we consider $\beta = 90°$ but vary γ from 90° to 80°. In this case, as expected, the light curve smoothly approaches a horizontal line and only about 9% of all observers would see anything but a featureless continuum. In Figure 7 we consider the case $\beta = \gamma = x$ varying from 10° to 90°. Since $\omega t = 0$ always corresponds to looking directly along the polar cap axis, the amplitude at 0° is independent of x but the width of the feature clearly increases with x. Furthermore, for x smaller than 85° the peak at 180° essentially disappears and the light curve goes from two peaks to one peak per cycle.

It is possible that the physical conditions existing in the polar cap region will alter photon emission as in the extreme case of radio pulsars. Accordingly we have considered beamed emission near Lambert's law. In Figure 8 we show the light curve of one 5° polar cap where $f(\delta) = \cos\delta$, and in Figure 9 the resulting two cap light curves. In this case the preferential emission of photons in the radial direction strongly biases the $\theta_0 = 0°$ position over the 180° position for the one cap curves, producing two cap curves that peak at 0° for all $R/2M$ and never flatten. The peak at $R/2M = 1.6$ remains but is much weaker because of the relative paucity of near tangential photons.

In Figure 10 we show the one cap light curves and in Figure 11 the two cap curves for $f(\delta) = \sin\delta$. As expected, the photon enhancement near the tangential plane has produced a much enhanced spike at 180° in the one cap curve. We note in Figure 9 that for $M = 0$, geometry and $\sin\delta$ beaming combine to produce a peak in the light at 45°, implying a four-peaked light curve. For more relativistic configurations, however, gravity acts to diminish near face-on brightness, shifting the

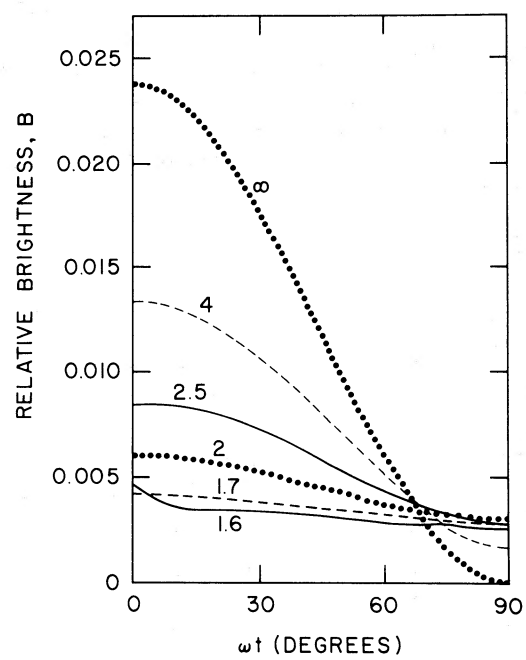

FIG. 9.—Relative brightness B (two polar caps) vs. ωt for $\alpha = 5°$, $f(\delta) = \cos\delta$, $\beta = \gamma = 90°$. Numbers denote values of $R/2M$.

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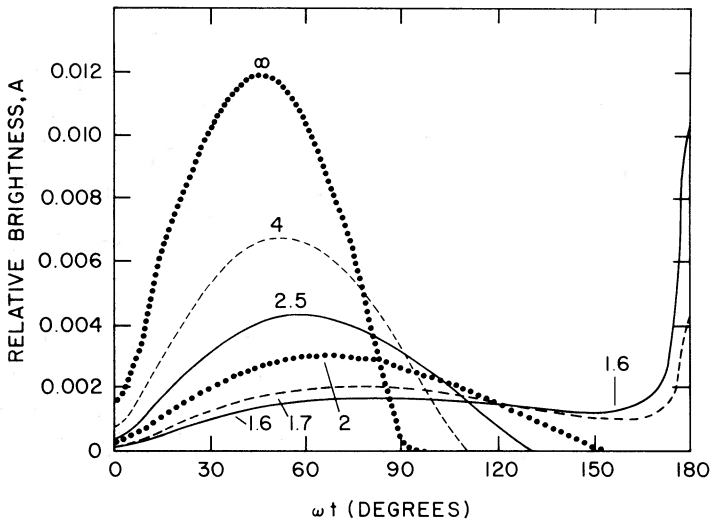

FIG. 10.—Relative brightness A (one polar cap) vs. ωt for $\alpha = 5°$, $f(\delta) = \sin\delta$, $\beta = \gamma = 90°$. Numbers denote values of $R/2M$.

peak to 90° and giving a two-peaked light curve. The peak still appears for the very relativistic case, but the continuum is no longer flat, owing to a relative dip near zero in the one cap case. These light curves in particular should be compared with those of WW's Figure 4 since for $\sin\delta$ beaming the tangential photon enhancement magnifies the effects of gravity.

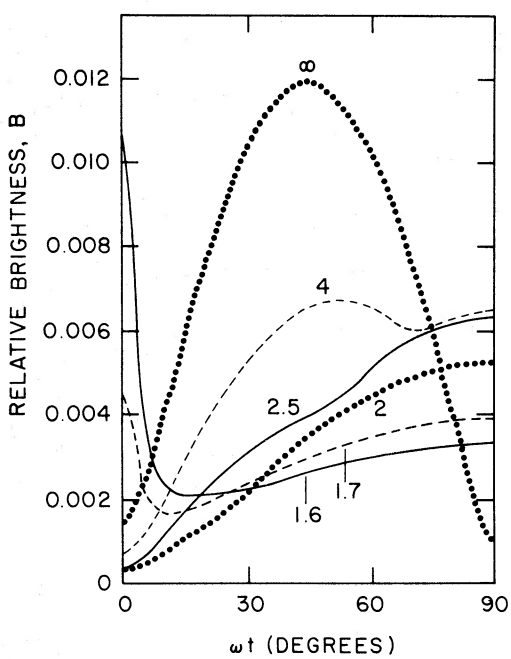

FIG. 11.—Relative brightness B (two polar caps) vs. ωt for $\alpha = 5°$, $f(\delta) = \sin\delta$, $\beta = \gamma = 90°$. Numbers denote values of $R/2M$.

VI. CONCLUSIONS

We have demonstrated that gravity can play a significant role in determining the light curves expected from neutron stars with hot polar caps. In general we find:

1. For thermal emission, gravity tends to flatten the light curve for $R/2M > 2$, giving very little observed variation for nominal pulsar values.

2. For $R/2M < 2$, a new feature appears in the form of a spike at 0° whose width is about the angular diameter of the polar cap and whose relative height increases with decreasing polar cap radius. This feature arises from emission in the vicinity of $\theta_0 = 180°$ and is the only structure in an otherwise featureless continuum.

3. If photon emission is caused to deviate from Lambert's law, gravity will alter the structure of the light curve but its variability is generally preserved.

Our results are applicable to the structure of X-ray pulsars in binary systems where accretion is assumed to give rise to hot polar regions. That the X-ray light curve arises simply from rotational eclipsing of these caps, as concluded by WW for certain systems, is ruled out by our work. In particular the conclusions reached by WW requiring $\sin\delta$ beaming should be reexamined in the light of the current work. It is probable that one or more of our model assumptions must be violated in nature. One interesting possibility is that the emission originates above the neutron star surface. While it might be argued in this case that gravitational effects are less important, it is actually *easier* to see the emitting region above the star since the required light bending is less severe. We can illustrate this by considering the minimum height above the stellar surface necessary such that a point at $\theta = 180°$ is visible to the observer at infinity. This condition guarantees that any emitting region, regardless of shape and size, will always be in view. From the

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equations of § II this height can be found for $R/2M = 2.5$ as 0.19 stellar radii. Once $\theta = 180°$ becomes visible to the observer, a phenomenon like that associated with the very relativistic configurations should occur and it may well be that the "eclipsed" region is that responsible for the X-ray peaks. This is currently under investigation. It is important to emphasize that no point that is more than 2 or 3 km above the surface of a typical neutron star is ever out of sight.

One further consequence of the flatter light curves predicted by relativity is that the integrated emission over one cycle is about 60% greater than that predicted from flat-space calculations adjusted by appropriate redshift factors. This factor is a minimum for $\beta = \gamma = 90°$ since the relativistic light curve is flatter and essentially independent of β and γ while the flat-space curve can only decrease in amplitude as β and γ are varied. Averaging over all β and γ would probably raise the difference closer to a factor of 2.

The sharply peaked feature that appears in the light curves of the most relativistic configurations is a clear demonstration that gravity acting alone is capable of producing a focused photon beam. While it is admittedly limited in the current work to extreme cases, it constitutes a very simple model for producing tightly collimated beams.

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APPENDIX

SURFACE BRIGHTNESS AND THE LIGHT CURVE OF A "FLASHING" SPHERE

In this appendix we show that the surface brightness (as seen by a stationary observer) of a homogeneously and isotropically radiating stationary sphere is a constant. This result actually follows from the fact that I_ν/ν^3 is a general relativistic scalar and constant along a photon path (Bardeen 1973). Here I_ν is the intensity per unit frequency interval. We will prove the constancy of the surface brightness in some detail, however, in order to develop certain results necessary to correct a calculation by Matese and Whitmire (1978) of the light curve of a "flashing" Schwarzschild sphere.

I. THE SURFACE BRIGHTNESS OF A SCHWARZSCHILD SPHERE

We assume that in the Schwarzschild metric produced by a sphere of mass M and radius R, there is a small photon detector with its center at $r = r_0$, $\theta = \pi/2$, $\phi = 0$ (with $r_0 \to \infty$), and with the surface of the detector perpendicular to the "x-axis." A photon emitted from the surface at the point P with coordinates $r = R$, $\theta = \pi/2 - \Phi_R$, $\phi = 0$, will hit the center of the detector (on the "x-axis") if its path is in the "x-z plane" and if the angle between the surface normal and the photon's initial direction of motion is δ. The impact parameter of this photon is denoted by b. If any one of the three quantities b, δ, and Φ_R is known, the other two are determined. [Note: $\Phi_R(b)$ is given by eq. (2.12), where $\Delta\phi$ corresponds to Φ_R.]

We now consider a "cone" of photons, all of which start at the point P; the half-angle of this cone is denoted by α, and the axis of the cone is in the direction of the "central" photon: the one which will hit the center of the detector. We assume that the detector is sufficiently large that every photon inside the cone will strike the detector.

For a photon on the surface of the cone, we let δ' denote the angle between the photon direction and the surface normal, while b' denotes the impact parameter. Each photon on the surface of the cone will be described by an angle ξ, defined so that when $\xi = 0$ or π, the orbit is in the "x-z plane"; when $\xi = 0$, $\delta' = \delta + \alpha$; when $\xi = \pi$, $\delta' = \delta - \alpha$; and when $0 < \xi < \pi$, the photon direction has a positive "y-component."

We found previously (eq. [3.11]) that

$$b = R(1 - 2M/R)^{-1/2} \sin\delta . \tag{A1}$$

Similarly,

$$b' = R(1 - 2M/R)^{-1/2} \sin\delta' . \tag{A2}$$

From the definition of ξ, it follows that

$$\cos\delta' = \cos\delta\cos\alpha - \sin\delta\sin\alpha\cos\xi . \tag{A3}$$

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We will now find the "plane" of the orbit of a photon at the surface of the cone, as a function of ξ. We will let ζ denote the angle between the "plane" of the orbit and the "x-z plane." The calculation will be performed as if spacetime were flat, but the result will be valid even in curved spacetime.

The angle ζ will be defined in such a way that for $\alpha \ll 1$, $\tan\zeta$ and $\sin\xi$ have the same sign. It is necessary to find ζ such that

$$\hat{y}\cos\zeta + (\hat{z}\cos\Phi_R - \hat{x}\sin\Phi_R)\sin\zeta$$

is perpendicular to

$$\hat{x}\cos(\Phi_R - \delta) + \hat{z}\sin(\Phi_R - \delta) + \{[\hat{x}\sin(\Phi_R - \delta) - \hat{z}\cos(\Phi_R - \delta)]\cos\xi + \hat{y}\sin\xi\}\tan\alpha. \tag{A4}$$

We obtain

$$\tan\zeta = \sin\xi\tan\alpha/(\sin\delta + \cos\xi\tan\alpha\cos\delta). \tag{A5}$$

For $\alpha \ll 1$,

$$\zeta \approx \alpha\sin\xi\csc\delta. \tag{A6}$$

By using equations (A2), (A3), and (A6), one can find b' and ζ for any photon (characterized by ξ) at the surface of the cone.

We are interested in determining the point where each photon hits the detector. This will enable us to find the area A_d of the part of the detector which intercepts the photons in the cone.

Now let $\Phi'_R \equiv \Phi_R(b')$. If we know b' for a photon, we can find Φ'_R. If we know ζ and Φ'_R, we can find the point where the photon strikes the detector. This is accomplished by considering a rotated coordinate system (with primed coordinates), such that

$$y = y', \qquad z = z'\sin\Phi_R - x'\cos\Phi_R. \tag{A7}$$

(We will not need to know x in terms of x' and z'.)

We also define rotated spherical coordinates:

$$x' = r\sin\theta'\cos\Phi', \qquad y' = r\sin\theta'\sin\Phi', \qquad z' = r\cos\theta'. \tag{A8}$$

These rotated coordinates are useful because

$$\Phi'_f = \zeta \quad \text{and} \quad \theta'_f = \Phi'_R, \tag{A9}$$

where Φ'_f and θ'_f are final coordinates of the photon, when it strikes the detector. From equations (A7)–(A9), and setting $r = r_0$, we obtain

$$y_f = r_0\sin\Phi'_R\sin\zeta, \qquad z_f = r_0(\cos\Phi'_R\sin\Phi_R - \sin\Phi'_R\cos\zeta\cos\Phi_R). \tag{A10}$$

We were able to do this part of the calculation as if spacetime were flat, because we are only interested in $r = r_0 \to \infty$.

In expressing y_f and z_f, the final coordinates of the photon, in terms of ξ, we use the equation

$$\begin{aligned}\Phi'_R &\approx \Phi_R + (b' - b)\, d\Phi_R/db \\ &\approx \Phi_R + R(1 - 2M/R)^{-1/2}\alpha\cos\delta\cos\xi\frac{d\Phi_R}{db}.\end{aligned} \tag{A11}$$

The result is

$$y_f = \alpha r_0\sin\Phi_R\csc\delta\sin\xi, \qquad z_f = -\alpha r_0 R(1 - 2M/R)^{-1/2}\frac{d\Phi_R}{db}\cos\delta\cos\xi, \tag{A12}$$

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to first order in α. Equation (A12) represents an ellipse whose area is given by

$$A_d = \pi\alpha^2 r_0^2 R(1-2M/R)^{-1/2}\cot\delta\sin\Phi_R\frac{d\Phi_R}{db}. \tag{A13}$$

The solid angle at the surface of the sphere is

$$\Delta\Omega_s = \pi\alpha^2. \tag{A14}$$

The element of surface area of the sphere is

$$dS = 2\pi R^2\sin\Phi_R\, d\Phi_R, \tag{A15}$$

and the solid angle in the observer's sky is

$$d\Omega_0 = (2\pi/r_0^2)\, b\, db, \tag{A16}$$

from equations (3.2) and (3.6).

The brightness seen by the observer, when the sphere radiates "isotropically," is proportional to (Matese and Whitmire 1978, hereafter MW)

$$(\Delta\Omega_s/A_d)(dS/d\Omega_0)\cos\delta, \tag{A17}$$

where the factor of $\cos\delta$ comes from Lambert's law. Because of equation (A1), (A17) is a constant. Thus the disk seen by an observer at infinity appears uniformly bright.

II. CORRECTION OF THE RESULT OF MATESE AND WHITMIRE

MW calculated the light curve of a stationary sphere in the Schwarzschild metric which emits an instantaneous flash of light. In part of their calculation they assumed that an initially circular pencil remains circular in cross section, thus ignoring the tidal distortion due to the gravitational field. As we have shown in equation (A12), this cross section is actually elliptical. MW calculated the correct vertical semiaxis (our eq. [A12] for z_f) but then assumed a circular cross section. Here we correct their calculation, using a simpler method, since we may now assume that surface brightness is proportional to solid angle subtended in the observer's sky. From equation (2.2)

$$\frac{dr}{dt} = \left[1-b^2(1-2M/r)/r^2\right]^{1/2}(1-2M/r). \tag{A18}$$

Substituting $u = M/r$ into this equation, we obtain

$$\Delta t = M\int[Z(u)]^{-1}\left[1-(b/M)^2 Z(u)\right]^{-1/2} du, \tag{A19}$$

where

$$Z(u) = (1-2u)u^2. \tag{A20}$$

If the photon travels from $r = R$ to $r = \infty$, then Δt is infinite. However, the quantity τ, defined by

$$\tau = \Delta t(b) - \Delta t(b=0), \tag{A21}$$

is finite, and is given by

$$\tau = M\int_0^{M/R} du[Z(u)]^{-1}\left\{\left[1-(b/M)^2 Z(u)\right]^{-1/2}-1\right\}. \tag{A22}$$

The element of solid angle, $d\Omega'$, subtended in the observer's sky, is

$$d\Omega' = 2\pi\theta' d\theta' = (\pi/r_0^2)\, d(b^2) \tag{A23}$$

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from equation (3.6). Thus, the brightness is proportional to

$$I(\delta)\frac{d}{d\tau}(b^2), \tag{A24}$$

where $I(\delta)$ is the beaming function, and the relation between b and δ is given by equation (3.11). We find that

$$l/l_0 = (I/I_0)\left[(b^2)'/(b^2)'_0\right] \tag{A25}$$

with

$$l_0 = l(\tau = 0), \qquad I_0 = I(\delta = 0), \qquad (b^2)' = \frac{d}{d\tau}(b^2), \quad \text{and} \quad (b^2)'_0 = (b^2)'|_{b=0}.$$

For isotropic emission, $I(\delta)$ is constant, and equation (A25) becomes

$$l/l_0 = (M/R)\left\{\int_0^{M/R} du\left[1-(b/M)^2 Z(u)\right]^{-3/2}\right\}^{-1}, \tag{A26}$$

where $b = b(\tau)$ as given by equation (A22). From equation (A22) τ is a monotonically increasing function of b, while from (A26) l is a monotonically decreasing function of b. Thus l is a monotonically decreasing function of τ, with none of the peaks obtained by MW. MW's light curves go to zero when Φ_R passes through π, whereas ours do not.

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JEFFREY M. COHEN: Department of Physics, University of Pennsylvania, Philadelphia, PA 19104

CHRIST FTACLAS: Department of Astronomy and Astrophysics, University of Pennsylvania, Philadelphia, PA 19104

KAY R. PECHENICK: General Electric Company, Valley Forge Space Center, King of Prussia, PA 19406

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